EEE3094S 2026
Tutorial Test 1
Solutions

Name: ____________________

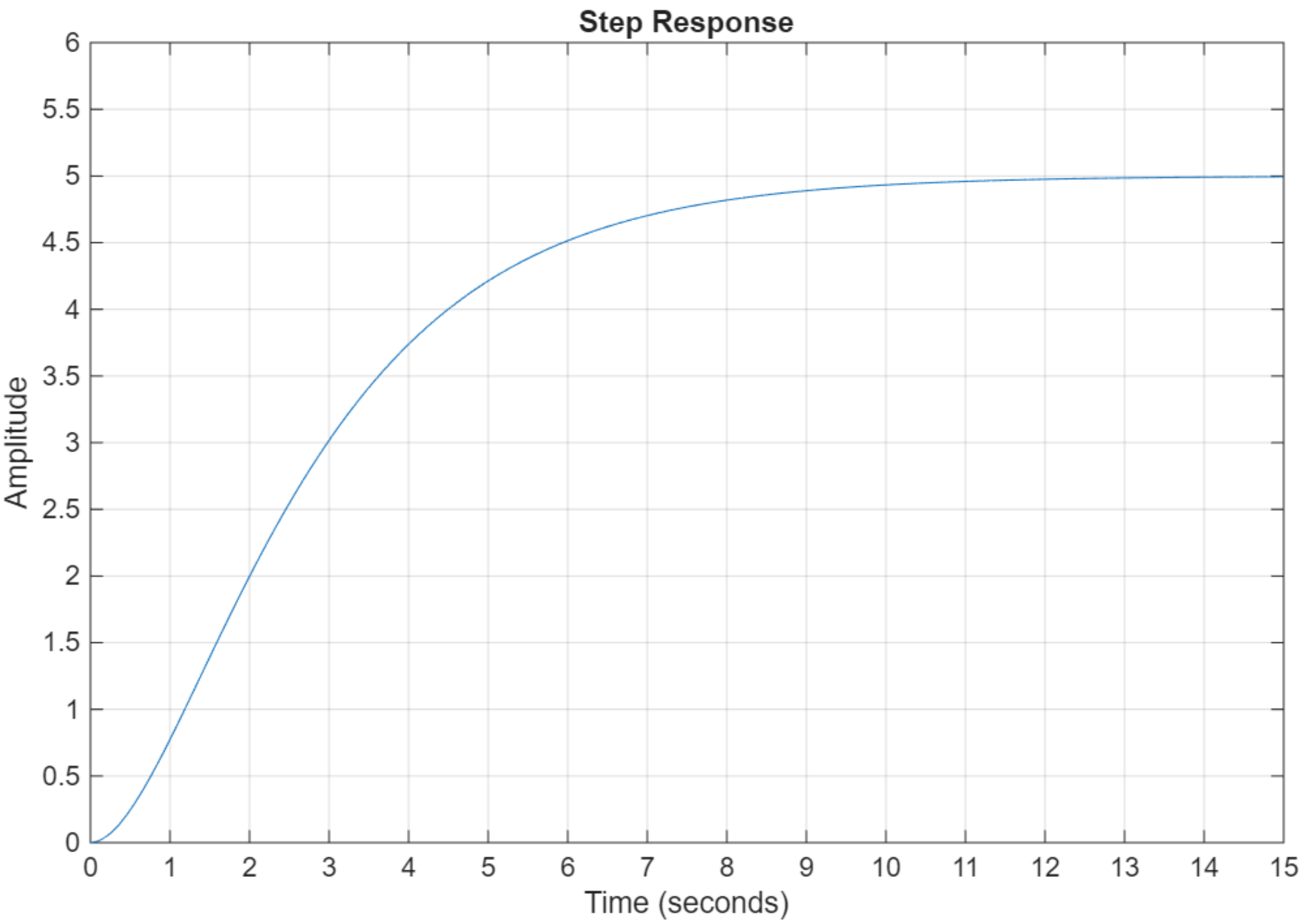
Student Number: ____________________

Instructions:

- The test is open book. You do not need to hand in your resources.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- You have 20 minutes to complete the test + 5 minutes of reading time.

1. Time domain system anaylysis (11 points)

Analyse the following unit step response for an unknown order system and answer the questions following.



a) (1 point) What is the gain of the system?

Solution: Unit step response, so the gain is the end value, hence 5. (1)

b) (4 points) This system is either first or second order. Which one do you suspect, and why? Argue your assumption by measuring typical metrics for these systems (hint: τ values).

Solution: While rather subtle, rise profile of the system showcases a delayed onset rise, different to the instant gradient of a normal first-order step response. (1) This indicates that the rise is second order. (1) This is reinforced by checking the rise time. If we assume a first order function, the time at which the system has risen to 63% of the final value shoud be half the time taken to reach within 86% of the final value. Inspecting the graph, the 63% point (3.15) is at 3.2s.(1) the value at 6.4s is 4.61, which is 92% of the final value.(1)

c) (2 points) Assuming no further information is available about the system, is it feasable to derive a transfer function by hand (i.e. excluding any fitting tools)? If not, what other information from the system would allow for modelling by hand?

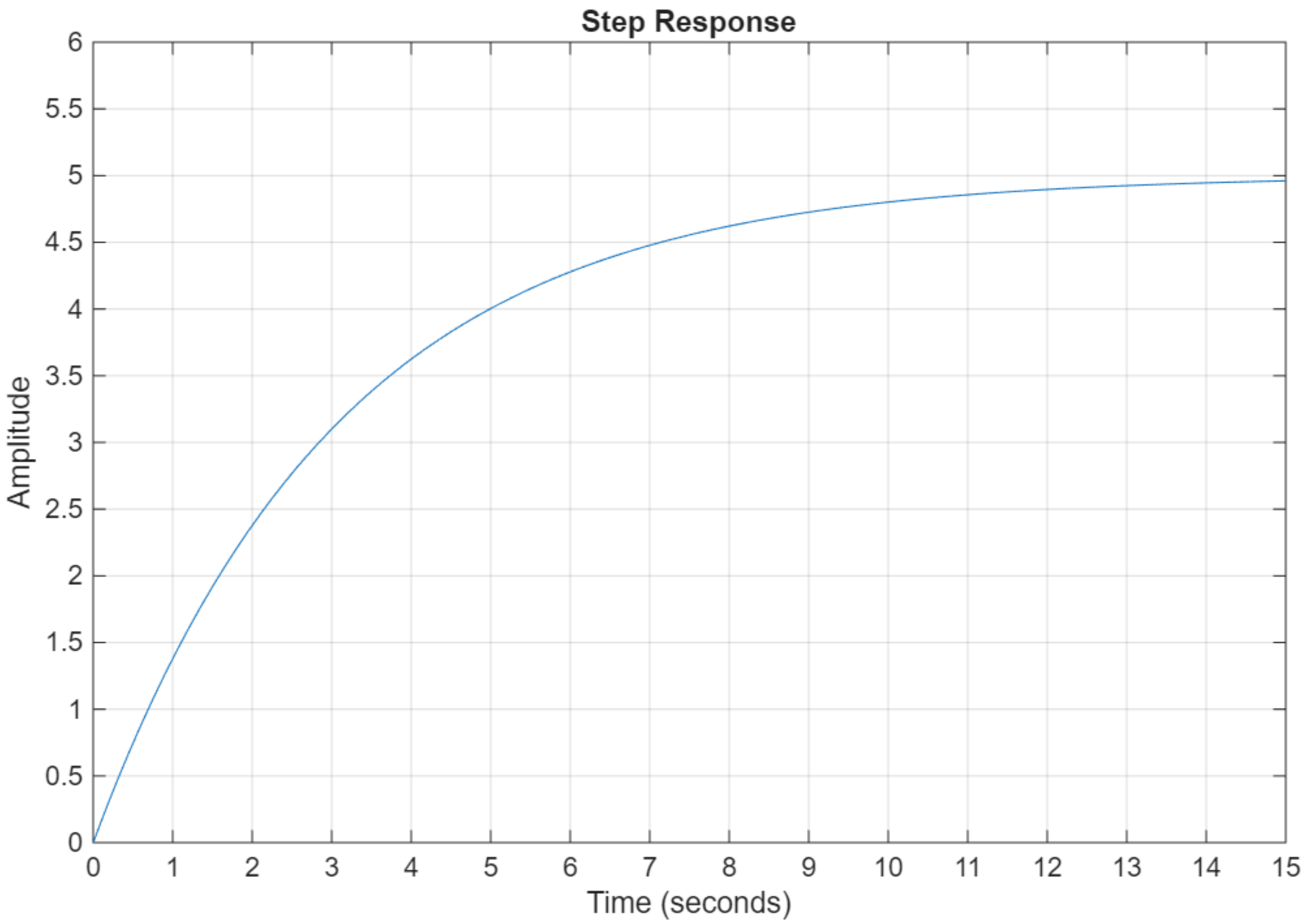
Solution: No, considering the system is second-order with two poles close to each other, it's not feasable to derive the transfer function by hand based only on the step response. (1) If the frequency response of the system was available, the system could be appropriately modelled.(1)

Note: Carry over error: If the answer to the previous Q was first order, then yes, we would have enough information (noting that this would then expose the fault in the first order assumption). Mark correctly if carry over error

d) (4 points) Irrespective of your answers to b) and c), suggest a first order model of the system, and sketch its unit step response on the plot below:

Solution: Using a standard 1st order response: $\frac{A}{\tau s+1}$, we know the gain A is 5. Estimating the time constant can be done in a variety of ways: any τ multiple from 1-5 will yield slightly different results. Taking the 1st multiple, i.e. the 63% point, we have $\tau = 3.1$s. This gives $P(s) = \frac{5}{3.1s+1}$, plot below: Marks:

1. τ (1)
2. Full transfer function (1)
3. Sketch: final value (1)
4. Sketch: shape and rise time (1)


Step Response
Amplitude
Time (seconds)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15